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Cloud Architecture Overview

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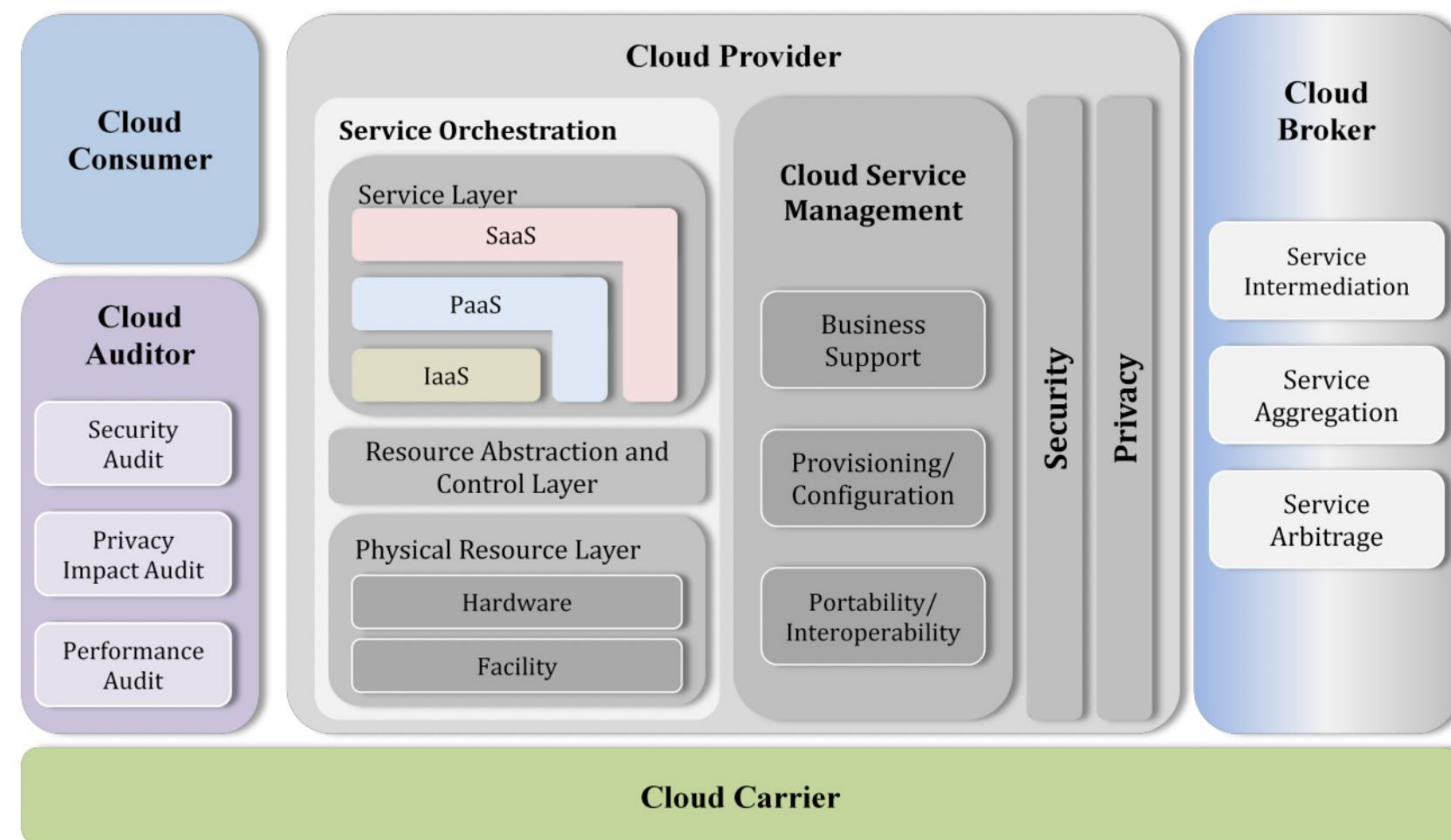
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- Management Layer
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- Security & Compliance Layer
- User & Business Layer

Introduction to NIST Cloud Computing Reference Architecture

- Developed by NIST to establish a vendor-neutral framework for cloud computing.
- Designed for U.S. Government and Federal CIOs, Procurement Officials, IT Managers.
- Key objectives:
 - Define components and taxonomy of cloud computing.
 - Enable secure, interoperable, and cost-effective cloud adoption.
 - Serve as a reference for standards and guidelines.



Key Actors

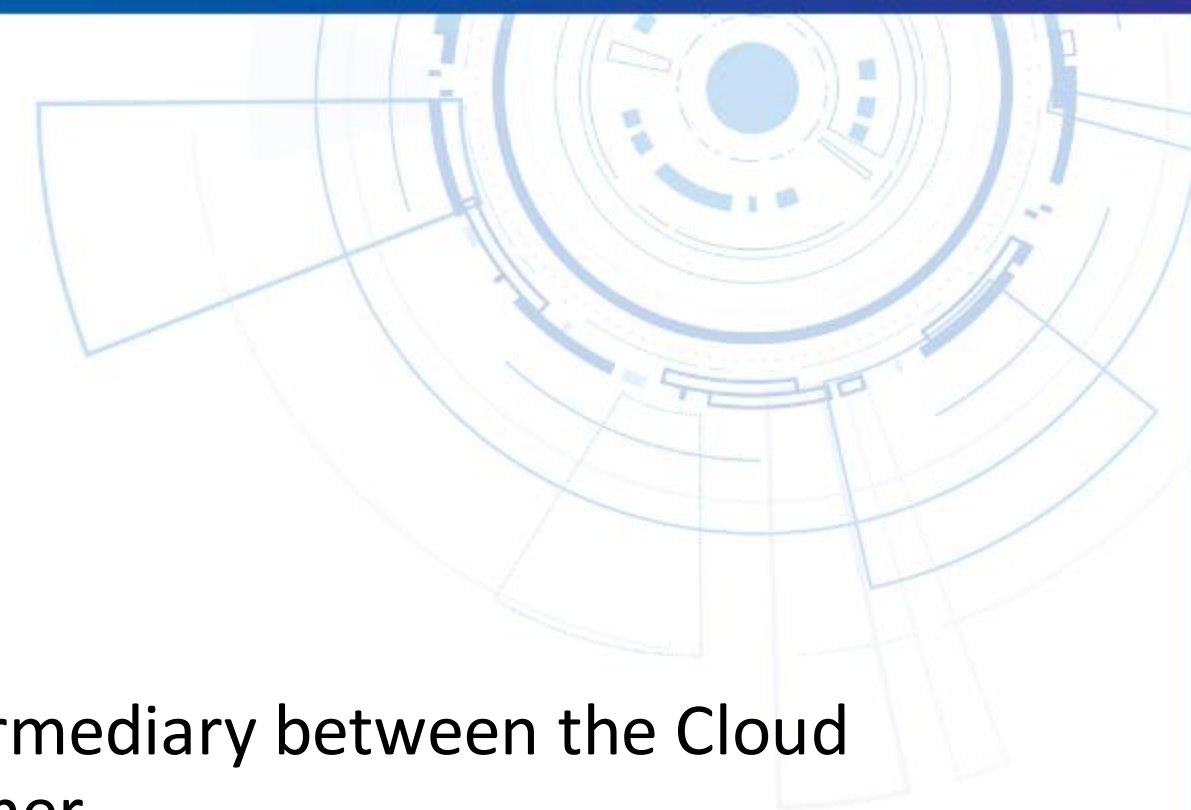
Cloud Consumer

- **Role:** Represents individuals or organizations that use cloud services.
- **Activities:**
 - Requesting services from a Cloud Provider or Broker.
 - Utilizing cloud resources such as SaaS, PaaS, or IaaS.
 - Interacting with SLAs (Service Level Agreements).

Cloud Provider

- **Core Role:** Delivers cloud services to consumers.
- **Key Functional Areas:**
 - Service Orchestration
 - Cloud Service Management
 - Security and Privacy

Key Actors



Cloud Auditor

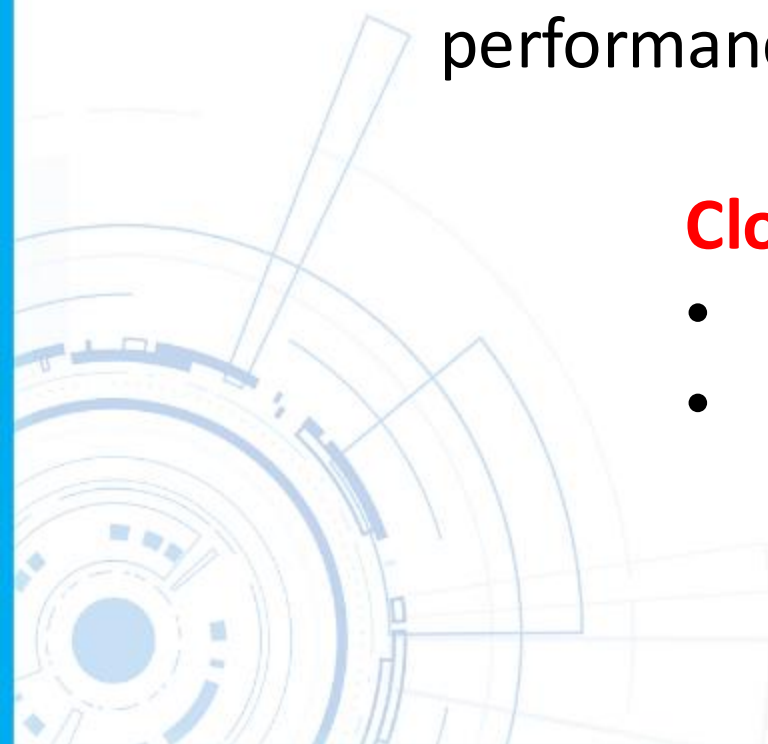
- **Role:** Performs independent assessments of cloud services.
- **Types of Audits:**
 - **Security Audit:** Verifies that security measures meet required standards.
 - **Privacy Impact Audit:** Ensures compliance with privacy regulations.
 - **Performance Audit:** Assesses service performance and reliability.

Cloud Broker

- **Role:** Acts as an intermediary between the Cloud Provider and Consumer.
- **Services Provided:**
 - **Service Intermediation:** Adds value by managing access or enhancing security.
 - **Service Aggregation:** Combines multiple cloud services into one.
 - **Service Arbitrage:** Dynamically selects services from multiple providers for flexibility.

Cloud Carrier

- **Role:** Facilitates the transport of cloud services from the provider to the consumer.
- **Responsibilities:**
 - Provides connectivity through networks and telecommunications.
 - Ensures secure and consistent data delivery.



Cloud Computing Technical Stack – Big Picture

1. Physical Infrastructure

- Data Centers
- Hardware
- Networks

3. Core Services

- Compute
- Storage
- Networking

5. Application

- Frontend Systems
- Backend Services
- APIs & Integration

7. User & Business

- Analytics & BI
- User Management
- Cost Management

2. Virtualization

- Compute Virtual.
- Storage Virtual.
- Network Virtual.

4. Management

- Orchestration
- Monitoring
- Automation

6. Security & Compliance

- Network Security
- Data Protection
- Identity Management

Physical Infrastructure Layer

Data Center

Power

- Main grid supply
- Backup generators
- UPS systems
- Cooling
 - HVAC systems
 - Hot/cold aisles
- Security
 - Access control
 - Surveillance
 - Environmental monitoring

Hardware

- Servers
 - Rack servers
 - Compute nodes
 - CPUs/GPUs
 - Memory (RAM)
- Storage
 - SSDs/HDDs
 - Storage arrays
 - Backup systems

Network

- Core Equipment
 - Switches
 - Routers
 - Load balancers
- Connectivity
 - Fiber cables
 - Network cables
 - Physical connections

Virtualization Layer

Virtualization is the enabler of Infrastructure-as-a-Service (IaaS) models in cloud computing. It allows cloud providers to offer scalable, on-demand resources to users without them needing to manage physical hardware directly.

Compute Virtualization:

- Virtual Machines (VMs): Allows multiple operating systems and applications to run on the same physical server, with each VM acting as an independent system.
- Hypervisors (e.g., VMware ESXi, Microsoft Hyper-V, KVM): Software that manages the creation and execution of VMs by abstracting the physical hardware.

Storage Virtualization:

- Abstracts physical storage resources (e.g., disks, SSDs) and combines them into a unified, flexible pool.
- Enables features like scalable storage allocation, snapshots, and replication.
- Examples: Storage Area Network (SAN), Network Attached Storage (NAS), or software-defined storage solutions.

Network Virtualization:

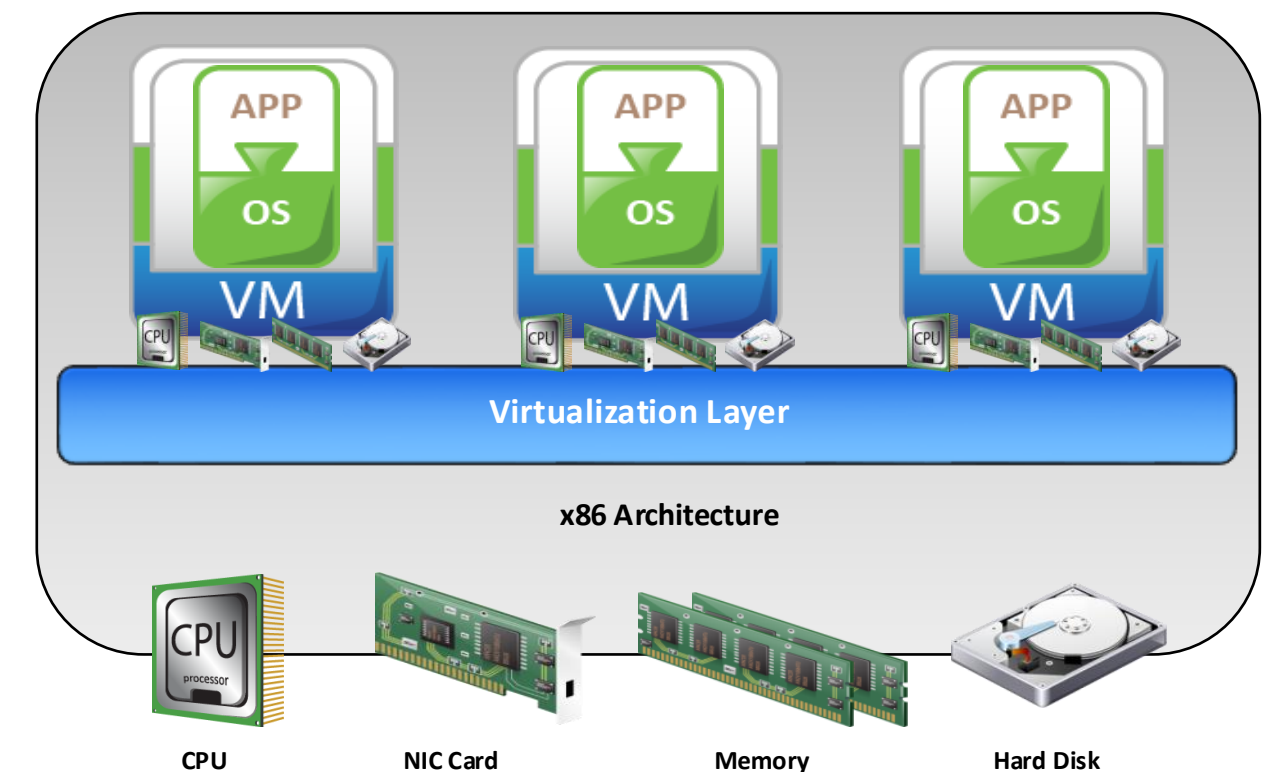
- Abstracts physical network components, allowing the creation of virtual networks that are decoupled from the underlying hardware.
- Includes features like virtual switches, virtual LANs (VLANs), and software-defined networking (SDN).
- Enables more flexible and scalable networking configurations.

Compute Virtualization

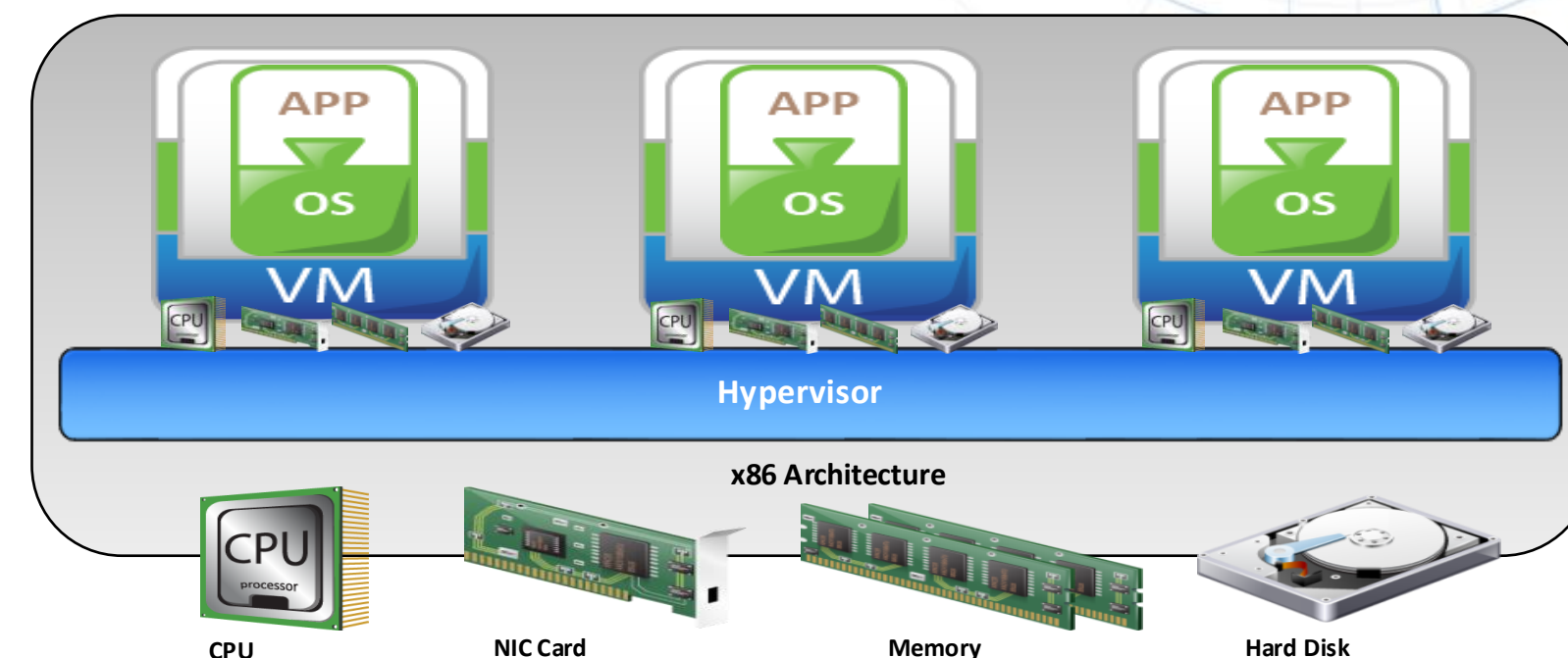
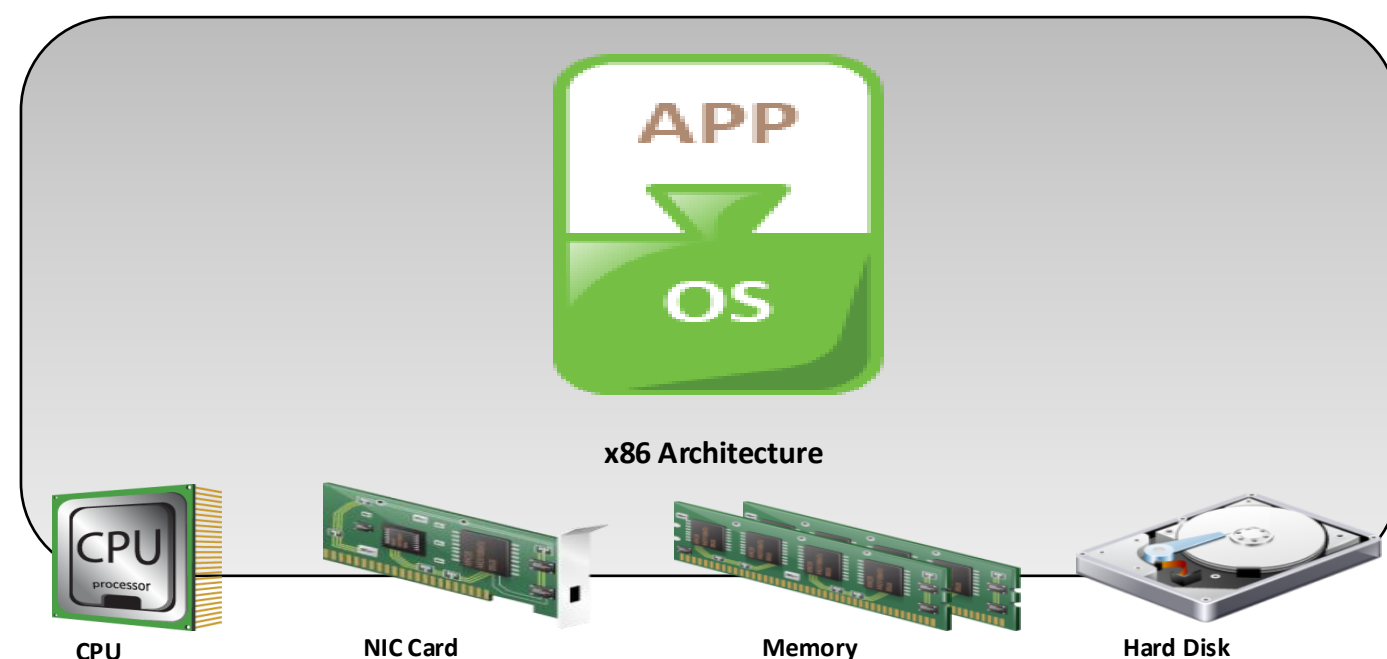
Compute Virtualization

It is a technique of masking or abstracting the physical compute hardware and enabling multiple operating systems (OSs) to run concurrently on a single or clustered physical machine(s).

- Enables creation of multiple virtual machines (VMs), each running an OS and application
 - VM is a logical entity that looks and behaves like physical machine
- Virtualization layer resides between hardware and VMs
 - Also known as hypervisor
- VMs are provided with standardized hardware resources



Compute Virtualization



Before Virtualization

- Runs single operating system (OS) per machine at a time
- Couples s/w and h/w tightly
- May create conflicts when multiple applications run on the same machine
- Underutilizes resources
- Is inflexible and expensive

After Virtualization

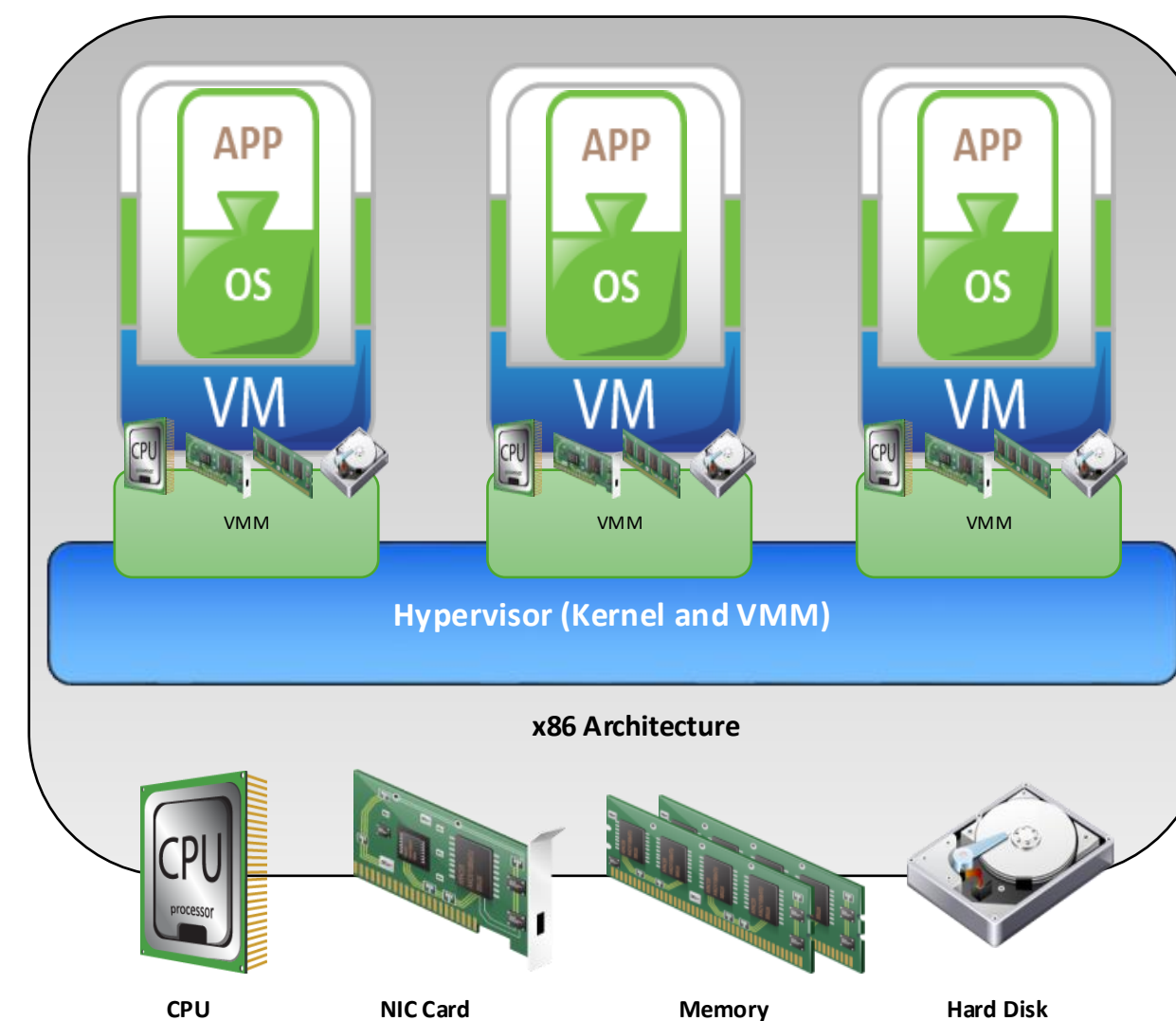
- Runs multiple operating systems (OSs) per machine concurrently
- Makes OS and applications h/w independent
- Isolates VM from each other, hence no conflict
- Improves resource utilization
- Offers flexible infrastructure at low cost

Hypervisor

Hypervisor

It is a software that allows multiple operating systems (OSs) to run concurrently on a physical machine and to interact directly with the physical hardware.

- Has two components
 - Kernel
 - Virtual Machine Monitor (VMM)



Storage virtualization

Storage virtualization

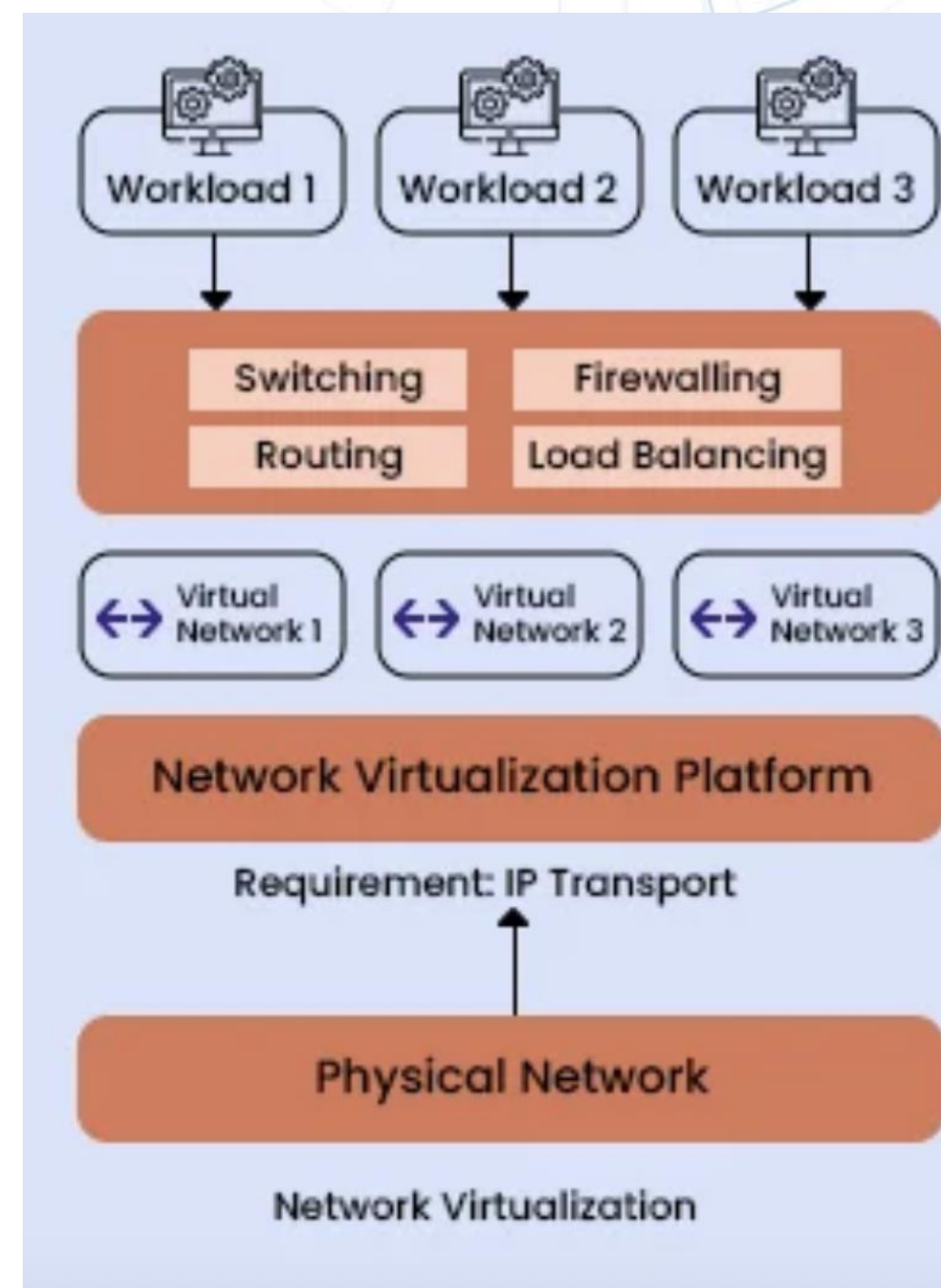
It is the process of masking the underlying complexity of physical storage resources and presenting the logical view of these resources to compute systems.

- Logical to physical storage mapping is performed by virtualization layer
- Virtualization layer abstracts the identity of physical storage devices



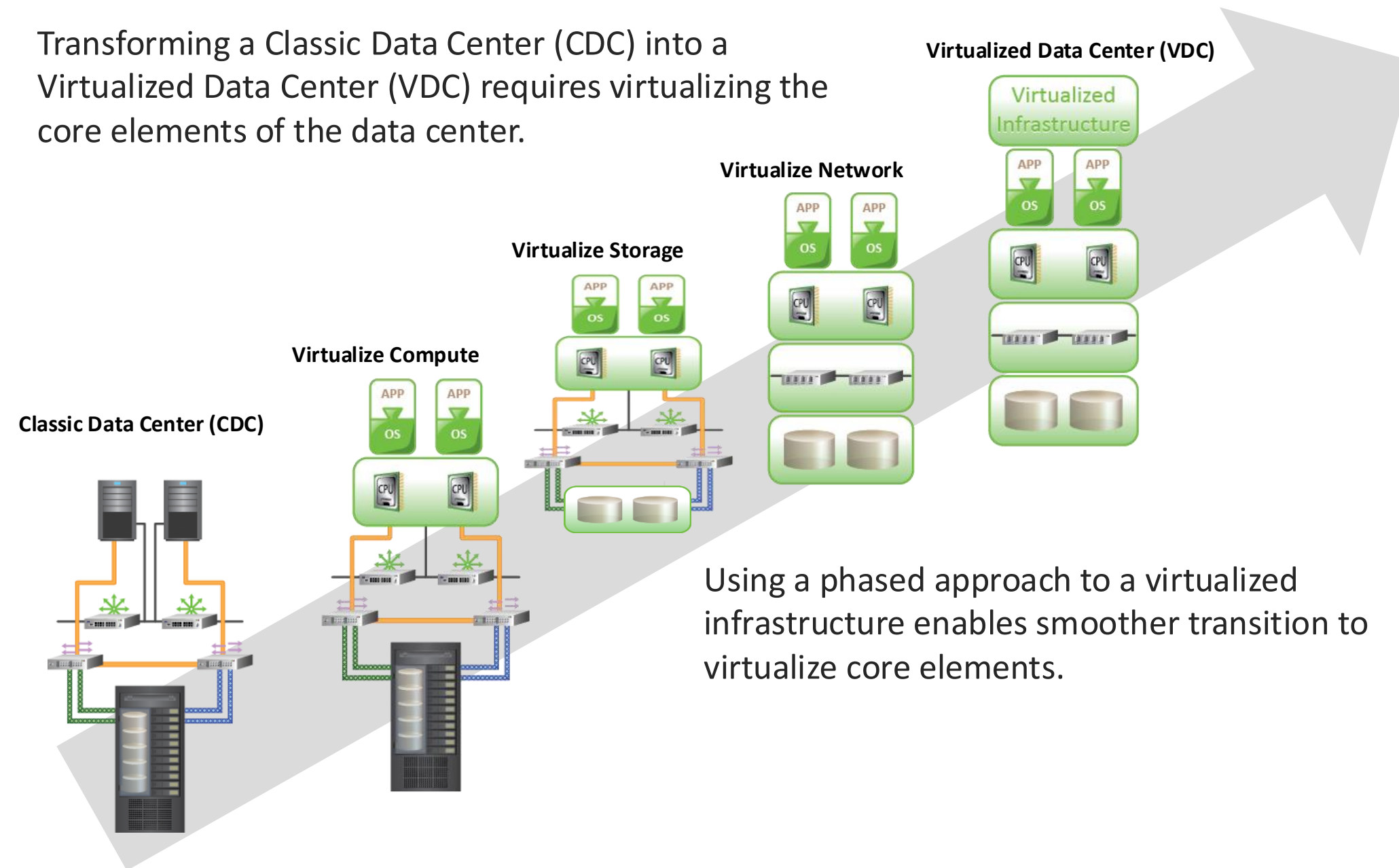
Network Virtualization

Network Virtualization is a technology that abstracts and combines hardware and software network resources into a single, manageable virtual network. It allows multiple virtual networks to share the same physical network infrastructure while maintaining isolation, flexibility, and efficiency. This is achieved by decoupling network functions, such as routing, switching, and security, from the underlying physical hardware.



Virtualized Data Center

Transforming a Classic Data Center (CDC) into a Virtualized Data Center (VDC) requires virtualizing the core elements of the data center.



Using a phased approach to a virtualized infrastructure enables smoother transition to virtualize core elements.

Core Services Layer

The **Core Services** layer is the heart of a cloud computing stack. It provides the essential building blocks required for running and managing applications, systems, and workloads in the cloud. These services form the foundation upon which other functionalities, such as application hosting, storage, and networking, are built.

Compute:

- Refers to the processing power needed to run applications, workloads, and services in the cloud.
- Examples include:
 - **Virtual Machines (VMs):** Virtualized compute instances that run operating systems and applications.
 - **Containers:** Lightweight, portable environments for applications (e.g., Docker, Kubernetes).
 - **Serverless Computing:** On-demand execution of code without managing servers (e.g., AWS Lambda, Azure Functions).
 - **Bare Metal Servers:** Physical servers offered without virtualization for high-performance needs.

Storage

- Provides the capacity to store, manage, and retrieve data in the cloud. It includes various types of storage for different use cases:
 - **Object Storage:** Ideal for unstructured data like files and multimedia (e.g., AWS S3, Google Cloud Storage).
 - **Block Storage:** Used for applications requiring low-latency access, like databases (e.g., AWS EBS, Azure Managed Disks).
 - **File Storage:** Network-based file systems (e.g., Azure Files, AWS FSx).
 - **Archival Storage:** Cost-effective long-term storage for infrequently accessed data (e.g., AWS Glacier).

Network

- Manages the connectivity between cloud resources and external systems. Networking services ensure data transfer, security, and availability:
- **Virtual Private Clouds (VPCs):** Isolated networks within a public cloud.
- **Load Balancers:** Distribute incoming traffic across multiple resources (e.g., AWS ELB, Azure Load Balancer).
- **Content Delivery Networks (CDNs):** Speed up content delivery by caching data closer to users (e.g., Cloudflare, AWS CloudFront).
- **Firewalls and Security Groups:** Protect resources by defining access control rules.
- **DNS Services:** Manage domain names and routing (e.g., AWS Route 53, Azure DNS).

Management Layer

- Focuses on the tools, processes, and frameworks required to efficiently operate, monitor, and optimize cloud resources. This layer plays a critical role in ensuring the smooth functioning of cloud environments by automating routine tasks, managing workloads, and providing insights for better decision-making.

Orchestration

- Refers to the automated coordination and management of cloud resources to ensure that workflows, applications, and services run seamlessly.
- Key functions:
 - Automating the deployment of applications and services.
 - Managing dependencies between services.
 - Ensuring that resources are provisioned and de-provisioned as needed.
- Tools: Kubernetes (for container orchestration), Apache Airflow, AWS CloudFormation, Terraform (for infrastructure as code).

Monitoring

- Provides real-time visibility into the performance, health, and usage of cloud resources.
- Ensures proactive detection and resolution of issues to minimize downtime and maintain performance.
- Key functions:
 - Tracking resource utilization (CPU, memory, storage).
 - Monitoring application performance (response times, error rates).
 - Generating alerts for anomalies or failures.
- Tools: Prometheus, Grafana, AWS CloudWatch, Azure Monitor, Datadog, New Relic

Automation

- Involves using scripts, tools, or platforms to handle repetitive tasks and processes automatically, reducing human intervention.
- Key functions:
 - Automating the scaling of resources based on demand (auto-scaling).
 - Automating backups and disaster recovery processes.
 - Managing configuration updates and patches.
- Tools: Ansible, Puppet, Chef, Jenkins, AWS Lambda.

The Application Layer

Refers to the components and services that allow developers and businesses to build, run, and manage applications in the cloud. This layer focuses on enabling the creation and operation of software that directly serves users, typically through frontend and backend systems, APIs, and integrations.

Frontend Systems

- Represents the user interface and experience (UI/UX) of the application.
- Includes web browsers, mobile applications, or desktop interfaces that interact with the underlying backend services.
- Technologies:
 - Web frameworks: React, Angular, Vue.js.
 - Mobile frameworks: Flutter, React Native.

Backend Services

- Handles the business logic, processing, and data management behind the scenes.
- Processes user requests, interacts with databases, and returns results to the frontend.
- Technologies:
 - Backend frameworks: Node.js, Django, Flask, Spring Boot.
 - Databases: MySQL, MongoDB, PostgreSQL, DynamoDB.

APIs & Integration

- Facilitates communication between different software components, systems, or services.
- Enables developers to integrate third-party services, connect microservices, or expose application functionality.
- Types of APIs:
 - RESTful APIs, GraphQL, gRPC.
- Tools: Postman (API testing), Swagger (API documentation).

Security & Compliance Layer

The **Security & Compliance** layer in the cloud computing technical stack is focused on safeguarding cloud environments, data, and applications while ensuring adherence to regulatory requirements and industry standards. This layer is critical to maintaining trust, protecting sensitive information, and ensuring that organizations operate securely within the cloud.

Network Security:

- Protects cloud-based networks and resources from unauthorized access, attacks, and breaches.
- Key Features:
 - Firewalls: Filter and control incoming and outgoing traffic (e.g., AWS Security Groups, Azure Firewall).
 - Virtual Private Networks (VPNs): Secure connections to cloud resources.
 - Intrusion Detection and Prevention Systems (IDS/IPS): Monitor and mitigate potential threats.

Data Protection:

- Ensures data confidentiality, integrity, and availability in the cloud.
- Key Features:
 - Data Encryption: Encrypts data at rest (e.g., AWS KMS, Azure Key Vault) and in transit (e.g., SSL/TLS).
 - Backup and Recovery: Ensures data can be restored in case of loss or corruption.
 - Access Controls: Defines who can access data and under what conditions.

Identity Management:

- Manages user identities and their access to cloud resources.
- Key Features:
 - Identity and Access Management (IAM): Manages roles, policies, and permissions (e.g., AWS IAM, Azure AD).
 - Single Sign-On (SSO): Enables users to access multiple applications with one set of credentials.
 - Multi-Factor Authentication (MFA): Adds an extra layer of security by requiring additional verification steps.

Security & Compliance Layer

Compliance Management

- Ensures that the organization meets industry regulations and standards.
- Common Compliance Standards:
 - GDPR (General Data Protection Regulation)
 - HIPAA (Health Insurance Portability and Accountability Act)
 - PCI DSS (Payment Card Industry Data Security Standard)
- Tools:
 - Cloud compliance tools like AWS Artifact, Azure Compliance Manager.

Threat Detection and Response

- Identifies and mitigates potential threats in real-time.
- Key Features:
 - Security Information and Event Management (SIEM): Collects and analyzes security data (e.g., Splunk, Azure Sentinel).
 - Automated Incident Response: Uses tools to automatically respond to detected threats.

Governance and Audit

- Ensures proper management of cloud resources and tracks activity to maintain accountability.
- Key Features:
 - Cloud Resource Policies: Define acceptable usage and behavior for resources.
 - Audit Logs: Track changes, accesses, and actions in the cloud (e.g., AWS CloudTrail, Azure Monitor).

User & Business Layer

Focuses on the interaction between cloud services and the end users or business operations. It ensures that cloud resources deliver value by enabling analytics, user management, cost management, and other business-oriented features. This layer is critical for aligning cloud services with organizational goals and user needs.

Analytics & Business Intelligence (BI):

- Provides tools and capabilities to analyze data stored or processed in the cloud to extract actionable insights.

•Features:

- Data visualization: Create charts and dashboards to make data insights easier to understand.
- Predictive analytics: Use AI/ML models to predict trends and patterns.
- Reports and KPIs: Generate reports to track business performance metrics.

•Examples:

- AWS QuickSight, Microsoft Power BI, Google Looker, Tableau.

User Management:

- Focuses on managing user accounts, access, and activities across cloud applications and services.

•Features:

- Role-based access control (RBAC): Assign permissions based on user roles.
- User activity tracking: Monitor and log user actions for auditing and compliance.
- Authentication and Single Sign-On (SSO): Simplifies user access across cloud services.

•Examples:

- Azure Active Directory, Okta, AWS IAM (Identity and Access Management).

Cost Management:

- Provides tools to monitor, control, and optimize the cost of cloud resources.

•Features:

- Billing dashboards: Track and analyze cloud spending.
- Budgeting and alerts: Set spending limits and receive alerts when thresholds are exceeded.
- Cost optimization recommendations: Identify unused or underutilized resources to save money.

•Examples:

- AWS Cost Explorer, Azure Cost Management, Google Cloud